



Basics of Quality and Patient Safety in Health Care

First year students



الهيئة العامة للاعتماد والرقابة الصحية

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Contents

No	Title	Page
1.	<i>Introduction of quality and safety</i>	8
2.	<i>Risk management and error reduction in healthcare</i>	14
3.	<i>Tips for Effective risk management</i>	23
4.	<i>Leadership for effective risk Management and Quality.</i>	26
5.	<i>Communication and collaboration for effective risk Management and Quality</i>	34
6.	<i>Evidence based practice for effective risk Management and Quality.</i>	42
7.	<i>Continuous quality improvement.</i>	52
8.	<i>Nursing informatics for quality improvement.</i>	62



University Vision

Al-Ryada University for Science and Technology aspires to be a distinguished and highly competitive leading university at the local and international levels in line with the sustainable development goals.

University Mission

The University is committed to providing a distinguished graduate capable of innovating and competing in the labor market and meeting the challenges of the future by providing a stimulating educational environment with intelligent and sophisticated techniques and strengthening partnerships with local and international universities and institutions and employing scientific research to achieve academic and research excellence while adhering to national identity, ethical values and community responsibility.



Faculty Vision

"The Faculty of Nursing at Al-Ryada University for Science and Technology aspires to be a leading institution in adopting distinguished holistic education that enhances students' personal capabilities and supports continuous self-development. This vision reflects positively on fair professional competitiveness at the local, regional, and international levels, and contributes to strengthening the academic, research, and societal advancement of the nursing profession."

Faculty Mission

"The Faculty is committed to employing modern educational methods and theories in alignment with scientific developments and quality-education standards, in order to graduate highly competent nursing professionals with strong competitive abilities in clinical practice and scientific research. The Faculty aims to contribute positively to the advancement of the profession, community service, and the achievement of sustainable development goals."



Introduction to quality and safety in healthcare

Overview of Quality and Safety in Healthcare

Quality and safety are fundamental components of healthcare, ensuring patients receive effective, efficient, and compassionate care while minimizing harm.

Definition of Quality: Doing the right thing at the right time in the right way to achieve the best possible results.

WHO definition of quality of care: it is the extent to which the health care services provided to individuals and patient populations improve desired health outcomes, to achieve this, health care must be safe, effective, timed, efficient, equitable and patient centered.

(Dimensions of quality):

1. Safe: delivering health care that minimizes risks, harm to service users including avoiding preventable injuries.
2. Effectiveness: Providing evidence-based care that achieves desired outcomes.
3. Efficiency: Optimizing resource utilization to minimize or avoid waste.
4. Timed: Delivering care promptly and reducing waiting times (reduce delay).
5. Equitable: - Delivering health care that doesn't differ in quality according to personal characteristics such as gender, race, ethnicity or socioeconomic status. Ensuring care is accessible, unbiased, and culturally sensitive.
6. Patient-centered & (value-based care): Focusing on individual patient needs, values, and preferences.



Safety, on the other hand, involves:

1. Preventing harm: Minimizing errors, adverse events, and injuries.
2. Mitigating risk: Identifying and addressing potential hazards.
3. Promoting a safety culture: Encouraging transparency, accountability, and continuous improvement.

Importance of Quality and Safety in Healthcare

Quality and safety are essential for several reasons:

1. Patient outcomes: High quality, safe care improves patient outcomes, reduces morbidity, and mortality.
2. Patient satisfaction: Quality care enhances patient satisfaction, trust, and loyalty.
3. Cost reduction: Preventing errors, reducing complications, and optimizing resource utilization decrease healthcare costs.
4. Professional reputation: Healthcare organizations and professionals' benefit from a reputation for delivering high-quality, safe care.
5. Regulatory compliance: Meeting quality and safety standards ensures compliance with regulatory requirement.

Key Quality and Safety Concepts:

1. Patient-centered care: Care that respects patients' values, preferences, and needs.
2. Evidence-based practice: Care that incorporates the best available research evidence.
3. Continuous quality improvement: Ongoing efforts to improve care processes and outcomes.
4. Safety culture: A culture that prioritizes safety, encourages reporting, and learns from errors.



5. Just culture: A culture that balances accountability with learning and improvement.

lecture questions:

Define the following:

1. Quality
2. WHO Definition: Quality of care.
3. Just culture

List

1. 5 Tenets of Quality of Care.
2. Importance of Quality and Safety in health care.
3. Initiatives for Accreditation

Quality improvement

The framework that is used to systematically improve the ways care is delivered to patients and this process has characteristics; measured, analyzed, controlled & improved.

Accreditation initiatives that integrate quality and safety standers include(non-profit):

1. The Joint Commission's International (JCI) accreditation standards, American initiatives focus on quality safety and patient centered care.
2. The Centers for Medicare and Medicaid Services' (CMS) quality initiatives emphasize quality, safety, and value-based care.
3. The Agency for Healthcare Research and Quality's (AHRQ) patient safety programs: focus on safety, quality and evidence-based practice.





4. The Institute for Healthcare Improvement's (IHI) quality and safety initiatives emphasize quality, safety, continuous improvement.
5. The Saudi Central Board for Accreditation of Healthcare Institutions **CEBAHI** Achieving CBAHI accreditation is mandatory for all healthcare facilities in Saudi Arabia.
5. The General Authority for Healthcare Accreditation and Regulation (GAHAR) Egyptian quality initiatives focuses on quality of care, patient safety and competency.

The relationship between quality and safety in healthcare.

*Quality and safety are **interdependent**, **intertwined**, with each influencing the other:*



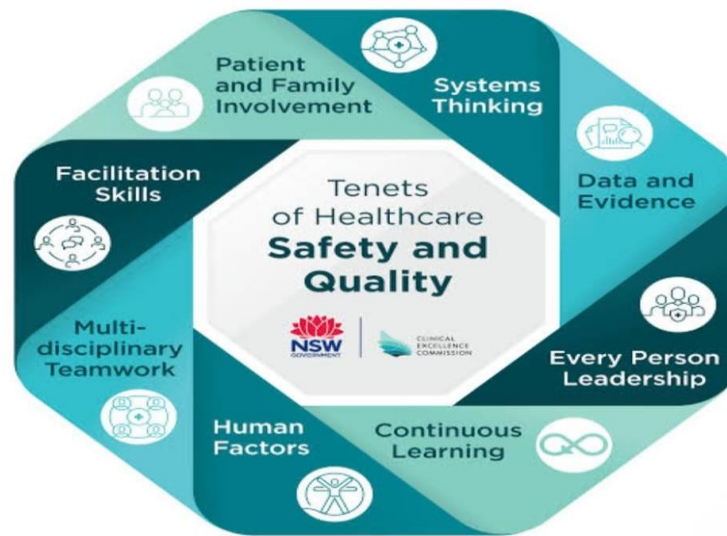
1. **Quality drives safety:** High-quality care reduces errors, adverse events, and harm.
2. **Safety enhances quality:** A safety-focused culture promotes transparency, accountability, and continuous improvement, ultimately enhancing quality.
3. **Shared goals:** Quality and safety share common goals, such as improving patient outcomes, reducing harm, and optimizing resource utilization.

Strategies for Improving Quality and Safety

1. Conduct regular quality and safety **assessments**: Identify areas for improvement.
2. **Implement** evidence-based practices: Use the best research available evidence.
3. Foster a **safety culture**: Encourage reporting, transparency, and accountability.
4. **Engage** patients and families: Involve them in care decisions and quality improvement (Multiciliary approach).

5. Provide **ongoing education and training**: Enhance staff knowledge and skills.

Tenets of Health care Quality and safety:



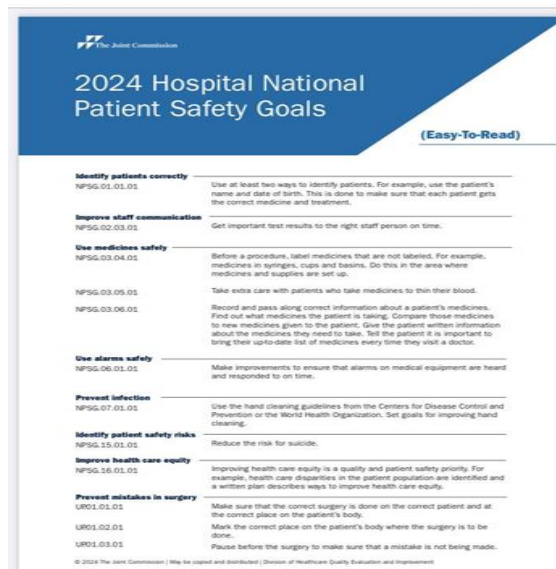
Challenges and Opportunities facing Quality and safety:

1. Complexity of healthcare systems: Multiple stakeholders, processes, and technologies.
2. Limited resources: Financial, personnel, and technological constraints.
3. Rapidly changing healthcare landscape: new technologies, policies, and reimbursement models.
4. Growing focus on value-based care: Emphasis on quality, safety, and cost-effectiveness.

Risk management and error reduction in healthcare

International patient safety goals (IPSGs):

- Identify patients correctly.
- Improve staff communication.
- Use medicine safely.
- Use alarms safely.
- Preventing infection
- Identify patient safety risks.
- Improve health care quality.
- Prevent mistakes in surgery.



Risk management and error reduction in healthcare:

Concepts of Risk and Error in Healthcare

1. **Risk:** Potential for harm or injury to patients, staff, or visitors.
2. **Error:** Unintended act or omission that can lead to harm or injury.
3. **Adverse event:** Unintended (undesirable but predictable) harm or injury resulting from medical care.
4. **Near miss:** Unintended event that could have resulted in harm or injury **but was prevented**.

Common Patient Safety Risks in Healthcare Settings:

1. **Medication Errors:** Wrong medication, dose, or administration route.
2. **Hospital-Acquired Infections (HAIs):** Infections contracted during hospital stay.
3. **Falls and Injuries:** Patient falls, slips, or trips.



4. **Surgical Site Infections (SSIs):** Infections at surgical sites.
5. **Venous Thromboembolism (VTE):** Blood clots in veins.
6. **Pressure Ulcers:** Skin damage due to prolonged pressure.
7. **Adverse Events:** Unintended harm or injury.
8. **Diagnostic Errors:** Incorrect or delayed diagnosis.
9. **Communication Breakdowns:** Inadequate or unclear communication.
10. **Equipment Failure:** Malfunctioning medical equipment.

Causes of Patient Safety Risks

1. **Human Factors:** Staff fatigue, stress, or lack of training.
2. **Systemic Issues:** Inadequate policies, procedures, or resources.
3. **Environmental Factors:** Poor lighting, noise, or ergonomic design.
4. **Technological Failures:** Equipment malfunction or software glitches.
5. **Consequences:** Patient harm, injury, or death; increased healthcare costs; decreased patient satisfaction.

Management: Management is the coordination and administration of tasks to achieve a goal, through the application of available resources.

Management is a process of getting work done through other people.

Benefits of Effective Risk Management

1. Improved patient safety: Reduce adverse events and near misses.
2. Reduced liability: Minimize risk of lawsuits and financial losses.
3. Enhanced reputation: Demonstrate commitment to patient safety and quality care.



Levels of Management:

The term “**Levels of Management**” refers to a line of demarcation between various managerial positions in an organization. The level of management determines a chain of command, the amount of authority & status enjoyed by any managerial position. The levels of management can be classified in three broad categories:

1. **Top level / Administrative level**
2. **Middle level / Executory**
3. **Low level / Supervisory / Operative / First-line managers**

Managers at all these levels perform different functions. The role of managers at all three levels is discussed below:

1. **Top Level of Management**
2. **Middle level / Executory**
3. **Low level / Supervisory / Operative / First-line managers**

Managers at all these levels perform different functions. The role of managers at all three levels is discussed below:



1. Top Level of Management

It consists of a board of directors, chief executive or managing director. They are sources of authority, and it manages goals and policies, they spend more time on planning.

The role of the top management can be summarized as follows:

1. Top management lays down the objectives and broad policies of the enterprise.
2. It issues necessary instructions for preparation of department budgets, procedures, schedules etc.
3. It prepares strategic plans & policies for the organization.
4. It appoints the executive for middle level i.e. departmental managers.
5. It controls & coordinates the activities of all the departments.
6. It is also responsible for maintaining contact with the outside world.
7. It provides guidance and direction.



2. Middle Level of Management

The branch managers and departmental managers constitute middle level.

They are responsible for the top management for the functioning of their department. In small organizations, there is only one layer of middle level management but in big organizations, there may be senior and junior middle level management.

Their role can be emphasized as -

- a. They execute the plans of the organization in accordance with the policies and directives of the top management.
- b. They participate in employment & training in lower-level management.
- c. They interpret and explain policies from top level management to lower level.
- d. They are responsible for coordinating the activities within the division or department.
- e. It also sends important reports and other important data to top level management.
- f. They are also responsible for inspiring lower-level managers towards better performance

3. Lower Level of Management

Lower level is also known as supervisory / operative level of management. It consists of supervisors, foreman, section officers, superintendent, they are concerned with direction and controlling function of management.

Their role includes:

- a. Assigning jobs and tasks to various workers.
- b. They guide and instruct workers for day-to-day activities.



- c. They are responsible for the quality as well as quantity of production.
- d. They are responsible for maintaining good relations in the organization.
- e. They supervise, guide the sub-ordinates and help to solve the complaints of the workers.
- f. They are responsible for providing training & motivation for the workers.
- g. They are the image builders of the organization because they are in direct contact with the workers.

Responsibilities of a Manager in risk management

To fulfill this role manager must possess:

- Budgeting.
- Planning
- Management theories
- Organizational structures
- Communication skills
- Rules and regulations
- Staff development
- Staff appraisal

Strategies for Managing Risk and Error in Healthcare Settings

1. Identify potential risks(assessment): Conduct risk assessments to identify potential hazards.
2. Develop policies, plans and procedures: Establish clear policies and procedures to mitigate risks.
3. Provide educational sessions and training: Educate staff on risk management and error reduction strategies.
4. Encourage reporting: Encourage staff to report near misses and adverse events.



5. Conduct analysis (root cause analyses): Investigate adverse events to identify underlying causes.
6. Implement corrective actions: Implement changes to prevent similar adverse events.

The risk management tools.

Risk Management Tools:

1. Failure Mode and Effects Analysis (FMEA): Identifies potential failures and their effects.

1. Identify potential failures
2. Evaluate the effects of each failure
3. Prioritize failures based on risk

2. Root Cause Analysis (RCA): Investigates adverse events to identify underlying causes.

1. Identify the adverse event
2. Gather data and information
3. Analyze the data to identify underlying causes
4. Implement corrective actions

3. SWOT Analysis: Identifies strengths, weaknesses, opportunities, and threats.

1. Identify internal strengths and weaknesses
2. Identify external opportunities and threats
3. Develop strategies to leverage strengths and opportunities and mitigate weaknesses and threats.

4. Decision Trees: Visualize decision-making processes to identify potential risks.

1. Identify the decision-making process



2. Create a visual representation of the process
3. Identify potential risks and opportunities

5. Fishbone Diagrams (Ishikawa Diagrams): Identify potential causes of problems.

1. Identify the problem
2. Create a visual representation of the problem
3. Identify potential causes

6. Pareto Analysis: Identify the most common problems or causes of problems.

1. Identify the problems or causes
2. Create a visual representation of the data
3. Identify the most common problems or causes

7. Risk Matrix: Evaluate and prioritize risks based on their likelihood and impact.

1. Identify potential risks
2. Evaluate the likelihood and impact of each risk
3. Prioritize risks based on their likelihood and impact

These risk management tools and techniques can help healthcare organizations identify, evaluate, and prioritize risks, and implement effective strategies to mitigate them.

Applying Risk Management Principles to Real-World Scenarios

1. Medication errors: Implement barcode scanning, automated dispensing systems, and double-checking processes.
2. Patient falls: Conduct fall risk assessments, implement fall prevention protocols, and provide education to staff and patients.



3. Surgical site infections: Implement evidence-based protocols for surgical site preparation, antibiotic prophylaxis, and post-operative care.
4. Medical equipment malfunctions: Regularly inspect and maintain medical equipment, provide education for staff on equipment use, and have backup equipment available.

Challenges and Opportunities

1. Limited resources: Allocate resources effectively to support risk management efforts.
2. Staff engagement: Encourage staff participation in risk management activities.
3. Changing healthcare landscape: Stay up-to-date with regulatory requirements and industry best practices.
4. Technological advancements: Leverage technology to support risk management efforts.

Tips for Effective risk management (Principles): عناوين فقط

I. Risk Assessment and Mitigation

1. Conduct Regular Risk Assessments: Identify potential safety risks and develop mitigation strategies.
2. Implement Evidence-Based Practices: use best practices and guidelines to reduce safety risks.
3. Use Safety Checklists and Tools: Utilize checklists and tools to ensure consistency and accuracy.

II. Leadership and Culture



1. Leaders are the architects of an organization's risk culture and are crucial in shaping how risks are perceived and handled. By demonstrating commitment to comprehensive risk management.
2. Leaders can inspire all levels of the organization to take informed risks and manage them.
3. Establish a Safety Culture: Encourage transparency, accountability, and open communication.
4. Set Safety Goals and Objectives: Develop and implement safety policies and procedures.
5. Provide Safety Training and Education: Educate staff on safety protocols and procedures.
6. The leader of a quality improvement team needs to deal effectively with a conflict between two units, it is best to appoint a facilitator to its membership.
7. Leaders are responsible for motivating quality culture.

III. Communication and Collaboration

1. Improve Communication: Enhance communication among healthcare teams, patients, and families.
2. Foster Collaboration: Encourage collaboration among healthcare professionals to ensure comprehensive care.
3. Use Technology to Enhance Communication: Leverage technology to facilitate communication and reduce errors.

IV. Patient Engagement and Empowerment

1. Encourage Patient Participation: Engage patients in their care and encourage them to speak up.
2. Provide Patient Education: Educate patients on safety protocols and procedures.



3. Support Patient Advocacy: Encourage patients to advocate for themselves and their loved ones.

V. Evidence based Practice: use the best researches to enhance patient care quality.

VI. Nursing informatics: and new technology in patient care and documentation

VII. Continuous Quality Improvement

1. Monitor and Analyze Safety Data: Track and analyze safety data to identify trends and areas for improvement.

2. Implement Quality Improvement Initiatives: Develop and implement initiatives to address safety concerns.

3. Evaluate and Refine Safety Protocols: Continuously evaluate and refine safety protocols to ensure effectiveness.

By implementing these strategies, healthcare organizations can reduce patient safety risks, improve quality of care, and enhance patient outcomes.

Lecture questions:

Define the following:

- a. Risk
- b. Error.
- c. Adverse Event
- d. Near miss

List:

- 1. International patient goals IPSGs.
- 2. 5 of Patient Safety Risks in Healthcare Settings.

3. Principles of Quality Improvement in Healthcare
4. Risk management tools.

Leadership for effective Risk Management & Quality Improvement

➤ **Definition of leadership:**

- ✚ Leadership is the ability of an individual or a group of individuals to influence and guide followers or other members of an organization.
- ✚ Leadership is the art of motivating a group of people to act toward achieving a common goal.



Characteristics of effective leader:

1. Effective Communicators

Leaders are excellent communicators, able to explain problems and solutions clearly and concisely. Leaders know when to talk and when to listen. In addition, leaders can communicate on different levels: one-on-one, via phone, email, etc.



2. Accountable and Responsible

Leaders hold themselves accountable and take responsibility for any mistakes. Leaders support and encourage individuality while abiding by organizational structure, rules, and policies that need to be followed.

3. Long-term Thinkers

Leaders are visionaries. This is evidenced by the leadership trait of being able to plan through concrete and quantifiable goals. They understand the need for continuous change and are open to trying new approaches to solve problems or improve processes.

4. Self-motivated

Leaders are self-motivated and can keep going and attain goals despite setbacks. In addition, good leaders try their best to exceed, not just meet, expectations.

5. Confident

Virtually all good leaders share the leadership trait of confidence. They can make tough decisions and lead with authority. By being confident, leaders can reassure and inspire others, establish open communications, and encourage teamwork.

6. People-oriented

Leaders are typically people-oriented and team players. They're able to foster a team culture, involve others in decision-making, and show concern for each team member. By being people-oriented, leaders can energize and motivate others. By making each individual feel important and vital to the team's success, they secure the best efforts from each member of the team.

7. Emotionally Stable

Leaders exercise good control and regulation over their own behavior and able to tolerate frustration and stress. Leaders can cope with changes in an environment without having an intense emotional reaction.



Types of Rewards in Quality Lead

- Advancement in career (promotion)
- Financial reward
- Appreciation
- Benefits as (health insurance, free gym, free parking....).
- challenge

Differences between Leader and Manager:

Basis	Manager	Leader
Origin	A person becomes a manager by virtue of his position.	A person becomes a leader based on his personal qualities.



Formal Rights	Manager has got formal rights in an organization because of his status.	Rights are not available to a leader.
Followers	The subordinates are the followers of managers.	The group of employees whom the leaders lead his followers.
Functions	A manager performs all five functions of management.	Leaders influence people to work willingly for group objectives.
Necessity	A manager is essential to a concern.	A leader is required to create cordial relations between a person working in and for organization.
Stability	It is more stable.	Leadership is temporary.
Mutual Relationship	<u>Not all managers are leaders.</u>	<u>Not all leaders are managers.</u>
Accountability	Manager is accountable for self and subordinates'	Leaders have no well-defined accountability.

	behavior and performance.	
Concern	A manager's concern is organizational goals.	A leader's concern is group goals and member's satisfaction.
Followers	People follow managers by virtue of job description.	People follow them on voluntary basis.
Role continuation	A manager can continue in the office till he performs his duties satisfactorily in congruence with organizational goals.	A leader can maintain his position only through day-to-day wishes of followers.

Leadership Styles for Quality and Safety

The six most common leadership styles are:

- Transformational Leadership.
- Delegative Leadership.
- Authoritative Leadership.
- Transactional Leadership.
- Participative Leadership.



- **Servant Leadership.**

1. Transformational leadership: Inspire and motivate staff to achieve quality and safety goals.

2. Servant leadership: Prioritize the needs of patients, families, and staff.

3. Collaborative leadership: Foster partnerships and collaborations to achieve quality and safety goals.

4. Emotional intelligence: Recognize and manage emotions to effectively lead quality and safety initiatives.

Leadership role in quality and risk management

- Strategic planning: Good leaders use proactive plans rather than reactive plans.
- Vision and Mission.
- Developing, implementing and maintaining risk management plan. Defining quality and safety goals.
- Engage stakeholders: Involve patients, families, staff, and other stakeholders in quality and safety initiatives.
- Establishing and ensuring culture of safety, Just Culture and quality culture.
- Establish quality committee.
- Developing and selecting the accrediting requirements.
- Defining quality and safety goals: Leaders establish clear goals and expectations for quality and safety initiatives.
- Allocating resources: Leaders ensure adequate resources are allocated to support quality and safety initiatives.
- Monitoring and evaluating progress: Leaders track progress, identify areas for improvement, and adjust as needed.

Strategies for Leading and Managing Quality and Safety Initiatives



1. Develop a quality and safety framework: Establish a structured approach to quality and safety initiatives.
2. Engage stakeholders: Involve patients, families, staff, and other stakeholders in quality and safety initiatives.
3. Foster a culture of transparency and accountability: Encourage open communication, reporting, and learning from errors.
4. Provide education and training: Ensure staff have the necessary knowledge and skills to support quality and safety initiatives.
5. Monitor and evaluate progress: Use data and metrics to track progress and identify areas for improvement.

Applying Leadership Principles to Real-World Scenarios

1. Leading a quality improvement initiative: Use leadership principles to guide a quality improvement initiative, such as reducing hospital-acquired infections.
2. Managing a patient safety incident: Apply leadership principles to manage a patient safety incident, such as a medication error.
3. Developing a quality and safety strategic plan: Use leadership principles to develop a strategic plan for quality and safety initiatives.

Challenges and Opportunities

1. Competing priorities: Balance quality and safety initiatives with other competing priorities.
2. Limited resources: Allocate limited resources effectively to support quality and safety initiatives.
3. Resistance to change: Overcome resistance to change from staff and other stakeholders.



4. Sustainability: Ensure quality and safety initiatives are sustainable over time.

Applying Leadership Principles for Patient Safety and Quality improvement:

1. Develop a quality and safety strategic plan: Use leadership principles to guide the development of a strategic plan.
2. Lead quality improvement initiatives: Apply leadership principles to guide quality improvement initiatives.
3. Manage patient safety incidents: Use leadership principles to manage patient safety incidents.
4. Foster a culture of quality and safety: Encourage a culture that prioritizes quality and safety.
5. Foster a just culture.

By applying these leadership principles, healthcare leaders can create a culture that prioritizes quality and safety, improves patient outcomes, and reduces harm.

Lecture questions

Define the following:

- a. Leadership
- b. Management.
- c. Adverse Event

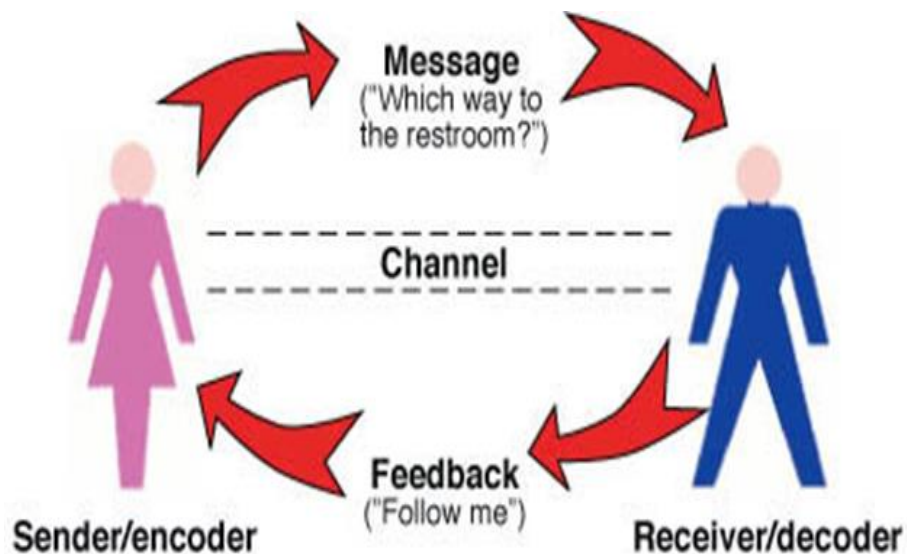
List:

1. Leadership traits.
2. Styles of leadership.
3. Role of Leaders in quality care

Effective Communication for Risk Management & Quality Improvement



Elements of communication process:



Importance of Effective Communication in Patient Safety

1. Reduces medical errors: Clear communication helps prevent misunderstandings that can lead to errors.



2. Improves patient outcomes: Effective communication ensures that patients receive timely and appropriate care.
3. Enhances patient satisfaction: Good communication helps patients feel informed, empowered, and satisfied with their care.
4. Supports teamwork: Communication facilitates collaboration among healthcare professionals, promoting a culture of safety.
5. Enhanced Team Collaboration: Fosters a culture of open communication and collaboration.

Standardized communication tools (Techniques):

1. **SBAR (Situation, Background, Assessment, Recommendation):** A structured communication framework for sharing critical information.
2. **Call-out:** A technique where team members explicitly state their intentions or actions to ensure clarity. One important aspect of a call-out is directing the information to a specific individual by name.
3. **Check-back:** A technique where team members confirm understanding by repeating back what they've heard.
4. **Debriefing:** A structured discussion after an event to review what happened, what went well, and what could be improved.
5. **Closed-Loop Communication:** A technique where the sender confirms that the receiver has understood the message.

Role of Effective Communication on Risk Management

➤ Early Detection of Issues:

Open communication channels allow staff to report potential risks or quality concerns promptly. This can prevent minor issues from turning into major problems.

- For example, a nurse noticing a pattern of **medication errors** can report it immediately, triggering an **investigation and corrective action**.



➤ **Error Prevention:**

Effective communication reduces the likelihood of errors caused by miscommunication or lack of information. This is particularly crucial during transitions of care, such as shift changes or patient transfers, where miscommunication can lead to serious consequences. Using standardized communication tools like SBAR (Situation, Background, Assessment, Recommendation) can greatly improve clarity.

➤ **Transparency and Accountability:**

Transparent communication fosters a culture of accountability, where staff members feel comfortable reporting errors and near misses without fear of reprisal.

This allows for a thorough analysis of incidents and the implementation of effective preventive measures.

➤ **Sharing Information:**

Clear and timely communication ensures that all relevant personnel have access to the information they need to provide safe and effective care.

This includes sharing new skills, new technologies and information as well as patient information, test results, changes in protocols, and updates on safety guidelines.

➤ **Patient Safety:**

Open communication with patients and their families is essential for ensuring patient safety. Communicate effectively with patients and families to empower them with knowledge about their care, encourage them to ask questions, express their concerns, and their feedback should be taken seriously.

Role of Collaboration on Risk Management:



➤ Multidisciplinary Teams:

Risk management and quality control **require the collaboration of various healthcare professionals, including physicians, nurses, pharmacists, administrators, and support staff.**

Multidisciplinary teams can bring diverse perspectives and expertise to the table, leading to more comprehensive and effective solutions.

➤ Shared Decision-Making:

Collaborative decision-making ensures that all stakeholders have a voice in the development and implementation of risk management and quality improvement initiatives.

This fosters a sense of ownership and increases the likelihood of successful implementation.

➤ Problem-Solving:

Collaboration **facilitates the identification and resolution of complex problems** that may contribute to risks or quality deficiencies.

By working together, teams can develop **innovative** solutions and implement them effectively.

➤ **Continuous Quality Improvement:**

Collaboration is **essential for continuous quality improvement**. By sharing data, best practices, new technology and lessons learned, healthcare organizations can continuously improve their performance and reduce risks.

For example, creating a quality improvement team that meets regularly to review patient outcomes and identify areas for improvement.

➤ **System Wide Improvements:**

Collaboration between different departments and even different healthcare facilities can create system-wide improvements. Sharing data, and best practices across a health system can greatly improve the overall quality of care.

Strategies for Improving Communication Among Healthcare Teams

1. Use clear and concise language: Avoid jargon and technical terms that may confuse team members.
2. Active listening: Pay attention to others, ask clarifying questions, and confirm understanding.
3. Standardize communication protocols: Establish consistent communication procedures, such as handoffs and debriefings.
4. Encourage open communication: Foster a culture where team members feel comfortable sharing concerns or questions.
5. Use technology to facilitate communication: Leverage tools like secure messaging apps, video conferencing, and electronic health records.

Applying Effective Communication Techniques to Real-World Scenarios



- **Critical situations:** Use clear and concise language during critical situations, such as code blues or rapid response team activations.
- **Regular safety meetings:** These meetings provide a forum for staff to discuss potential risks, share information about incidents, and develop preventive measures.
- **Standardized protocols:** during handoff implementing standardized protocols, and checklists for common procedures improves consistency, and therefore reduces risk.
- **Rounds and using different communication tools:** Encourage multidisciplinary teams to communicate effectively during rounds to coordinate care
- **Incident reporting systems:** These systems allow staff to report errors and near misses anonymously, promoting a culture of safety.
- **Simulation training:** Simulation training allows healthcare professionals to practice communication and collaboration skills in a safe and controlled environment.
- **Patient education.**

Barriers to Effective Communication

1. **Language barriers:** Differences in language or dialect can hinder communication.
2. **Cultural barriers:** Different cultural backgrounds can lead to misunderstandings.
3. **Hierarchical barriers:** Power imbalances can intimidate team members from speaking up.
4. **Technological barriers:** Poorly designed or malfunctioning communication tools can hinder effective communication.

Barriers to Effective Communication



Strategies to Overcome Barriers

1. **Language interpretation services:** Provide interpretation services to bridge language gaps.
2. **Cultural competency training:** Educate team members on cultural differences and nuances.
3. **Flattening hierarchies:** Encourage open communication and empower team members to speak up.
4. **Regular technology maintenance:** Ensure communication tools are functioning properly.

. Challenges to Implementing Effective Communication Techniques

1. **Time Constraints:** Healthcare teams often have limited time to communicate effectively.
2. **Information Overload:** Healthcare teams may be overwhelmed by the amount of information they need to communicate.
3. **Interdisciplinary Communication:** Communication breakdowns can occur between different healthcare disciplines.
4. **Technological Barriers:** Poorly designed or malfunctioning communication tools can hinder effective communication

Strategies to Overcome Challenges



1. **Prioritize Communication:** Make time for effective communication, even in busy healthcare environments.
 2. **Use Clear and Concise Language:** Avoid using jargon or technical terms that may confuse team members.
 3. **Implement Standardized Communication Protocols:** Establish consistent communication procedures to reduce errors.
 4. **Provide Regular Training and Feedback:** Educate healthcare teams on effective communication techniques and provide regular feedback.
- By implementing effective communication techniques and strategies, healthcare teams can improve patient safety, outcomes, and satisfaction.

Evidence Based Practice for Risk Management & Quality Improvement:

Introduction

Evidence-based practice (EBP) is the process of collecting, processing, and implementing best research findings to improve clinical practice, the work environment, or patient outcomes. According to the American Nurses Association (ANA), nursing interventions should be practical, methodical decisions based on EBP research studies. Utilizing the EBP approach to nursing practice helps us provide the highest quality and most cost-efficient patient care possible.

Definition of Evidence Based Practice:

Evidence-based practice is a systematic approach that emphasizes the use of **best evidence**, **clinical experience** and **patient preferences** to make

decisions about patient care and treatment that leads to the best patient outcomes, by selecting interventions that have the greatest chance of success.

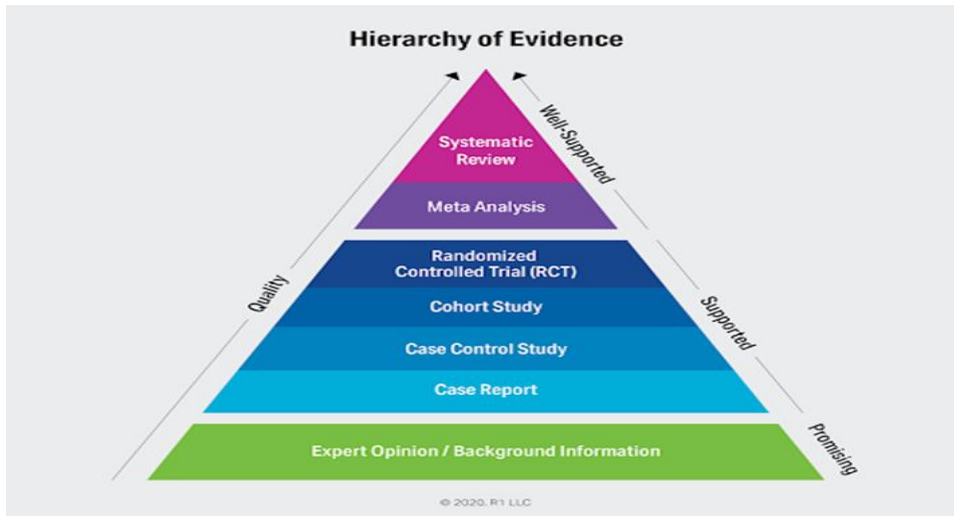
EBP is a process used to review, analyze and translate the latest scientific evidence.



EBP has **one** goal, **two** fundamental principles, **three** components and **five** steps.

- **One goal** is to improve the quality of clinical care through providing the most **effective** and **efficient care** possible. It also helps to **reduce variation in care** and improve **patient outcomes**.

- Two principles are hierarchy of evidence and insufficiency of evidence alone in decision-making.



- Three components are evidence, expertise and expectations of patients (triple Es).
- Five steps are asked, acquire, appraise, apply and assess (5 As).



Evidence Based Practice Process (5As)

The Components of Evidence Based Practice

The main definitions of EBP agree that there are **three** main components

(i) Best Evidence: evidence is simply **information** that supports a conclusion; in healthcare, this typically refers to the **effectiveness of interventions** in achieving specific outcomes.

(ii) Clinical Experience: It's the ability to use our clinical skills and experience to rapidly identify each patient's unique health state and diagnosis, their individual risks and benefits of potential interventions and their personal values and expectations.

(iii) Patient Preferences: the unique preference, concerns and expectations each patient brings to a clinical encounter, and which must be integrated into clinical decisions if they are to serve the patient.



The Components of Evidence Based Practice

Benefits of Evidence Based Practice:



Evidence-Based Practice has many benefits for patients, nurses, and health care organizations.

1- For patients:

1. Improved care and reduced time wasted on inappropriate treatments
2. Enhances consistency in care
3. Increases understanding of investigations and treatment
4. Increases accountability for care provision

2- For clinical Practitioners / nurse:

1. Actively involves them in determining the appropriateness and effectiveness of care
2. Help in redefining and/or changing practice where necessary
3. Enhances quality of care
4. Presents best evidence of practice to patients and carers
5. Increases confidence in practitioners

3- For the health care organization

1. Enables consistent decision making
2. Reduces variation in services
3. Promotes cost-effectiveness
4. Promotes integration of activities (e.g. research and development, clinical audit, continuing professional development)
5. Increase accountability to the public for the service provided

Five Key Steps of Evidence Based Practice (5As)

Step (1) Develop a researchable question (Ask):

Asking questions in a particular format is an important first step. It specifies the outcomes you are looking for. It also helps in assessing the relevance of the information to your patient's problem.

The **PICO** model is often used to define a problem and formulate a specific question: **P**opulation; **I**ntervention; **C**omparison; **O**utcome. An example of the PICO is as follows: In hospitalized patients over **60 years** of age, how effective is **a falls prevention** program in comparison to the **normal standard** of care in **decreasing falls and falls injury** rates?

⇒ In (**P**) patients aged 65 years and over how does (**I**) the use of an influenza vaccine compared to (**C**) not receiving the vaccine (**O**) influence their risk of developing pneumonia during the flu season?

Step (2) Select resources and conduct the search (Acquire):

- In this stage searching for evidence.
- Information is gathered from several sources including: systematic reviews, clinical practice guidelines and searching journal publications for pertinent research articles.
- **Sources of evidence for nursing practice**

Sources of evidence	Actions
Research	Search published and unpublished literature – original research (primary sources), reviews (secondary sources), clinical guidelines (e.g. NICE), published research-based standards of care Generate own evidence through research
Experience (professional or general)	Reflect on practice: articulate reflections, facilitate discussion, search the literature
Theoretical experience (not research-based)	Search the literature (published or unpublished), learn from others, facilitate discussions
Evidence gathered from patients and/or their carers	Search the literature for experiential writings and/or research findings, use audit data, levels of satisfaction and/or complaint, facilitate discussion and collaborative decision making
Evidence passed on from role models and/or experts	Facilitate discussion and observation, consult with experts. Search the literature for findings from consensus-reaching techniques (focus groups, Delphi surveys, nominal groups technique and consensus conferences)
Evidence derived from policy documents	Scrutinize documentation, facilitate discussion



Step (3) Review and appraise evidence (Appraise):

Assess the strength of the evidence. We need to know which form of evidence is ‘best’ and how we reach conclusions about its quality (as illustrated in **table 2**).

Critical Appraisal is the way to evaluate any research, **to be sure that:**

- **Validity:** The evidence is telling the truth.
- **Relevant:** The evidence should directly answer the question
- **Applicability:** Can the results be applied to local population.

Step (4) Applying the evidence (apply):

choose interventions and select the most valid evidence search and integrate the evidence with one's clinical expertise and the unique needs and values of target patient through:

- ▶ **GUIDLINE:** that collaboratively communicates regular reviews & updates
- ▶ **PATHWAY:** this describes how patient moves through his treatment
- ▶ **PROTOCOLS:** instructions for practitioners to safeguard patients as any deviation from it needs explanation and policies

Step (5) (Assess) Evaluate the outcome of evidence-based practice changes:

Determine how the practice change is used. Audit and feedback demonstrate the gap between actual and desired results and address questions such as:

- Did the clinicians receive the information about a practice change?
- Did they adhere to the practice change?
- How difficult or smooth was it to use the new way.



In this stage, evaluate the impact of EBP on patients, efficacy, efficiency, economic analysis, & health status impact.

When evaluating the impact, consider the following:

- The client's outcomes.
- Cost effectiveness.
- Client satisfaction.
- Health care providers satisfaction.

Application of evidence-based practices in real-world scenarios.

Real-World Scenario 1: Managing Hypertension

- Patient presents with hypertension (HTN)
 - Evidence-based guideline recommends lifestyle modifications and medication
 - Healthcare providers assess patient's needs, preferences, and values
 - use (**5As**) of **evidence-based practice to manage hypertension.**
- Patient and provider collaborate to develop a treatment plan incorporating lifestyle modifications and medication.

Real-World Scenario 3: Preventing Hospital-Acquired Infections

- A - Patient is admitted to the hospital.
- Evidence-based guideline recommends implementation of infection prevention and control measures
 - Healthcare provider assesses patient's risk factors for hospital-acquired infections
 - Patient and provider collaborate to implement infection prevention and control measures

B- Steps to Apply Evidence-Based Practices in Real-World Scenarios

1. Assess patient needs, preferences, and values



2. Retrieve relevant evidence-based guidelines and research
3. Appraise the evidence: Evaluate the quality, validity, and relevance of the evidence.
4. Integrate evidence into practice: Combine evidence with clinical expertise and patient preferences
5. Evaluate outcomes: Monitor and evaluate patient outcomes to refine evidence-based practices

Common Challenges in Applying Evidence-Based Practices

1. Limited access to evidence-based guidelines and research
2. Insufficient time or resources to implement evidence-based practices
3. Resistance to change: Healthcare providers may be hesitant to adopt new practices
4. Patient factors: Patient preferences, values, or circumstances may influence the application of evidence-based practices

Strategies to Overcome Challenges

1. Provide education and training: Educate healthcare providers on evidence-based practices and guidelines
2. Develop clinical decision support tools: Create tools to guide clinical decision-making
3. Implement policies and procedures: Develop policies and procedures to support evidence-based practices
4. Encourage interdisciplinary collaboration: Foster collaboration among healthcare professionals to promote evidence-based practices



By applying evidence-based practices in real-world scenarios, healthcare professionals can provide high-quality, patient-centered care that improves patient outcomes and reduces healthcare costs.

lecture questions:

Define the following:

- Evidence based practice
- Validity:
- Reliability
- Relevant:
- Applicability

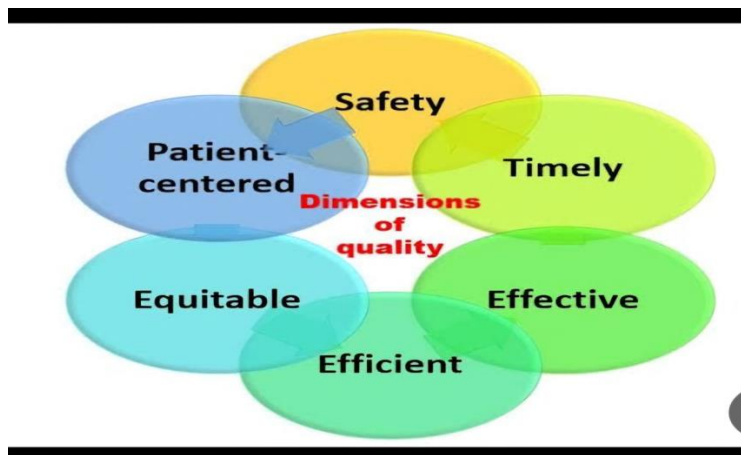
List :

- Three components of EBP.
- Five steps OF EBP.
- Benefits of Evidence Based Practice.
- Hierarchy of EBP

Continuous Quality Improvement principles and tools

Quality: *is doing the right thing, on the right time* within the *right way to* achieve best possible results.

Dimensions of Health care Quality according to (IOM):



1. **Quality Assurance (QA):** A proactive process focused on ensuring that healthcare services meet defined quality standards before problems occur. Goal: Prevent errors by establishing systems, protocols, and standards.
2. **Quality Control (QC) :**A reactive process used to identify and correct issues in healthcare delivery after they occur.Goal: Detect and fix deviations from quality standards.
- 3.**Quality Improvement (QI):** An ongoing, systematic process to analyze performance and make measurable improvements in healthcare processes and outcomes. Goal: Continuously enhance care delivery and patient outcomes. **All of them are called Total Quality Management**

Principles of Quality Improvement in Healthcare

1. **Patient-centered care:** Focus on patient needs, preferences, and values.
2. **Continuous improvement:** Regularly assess and improve care processes.
3. **Collaboration:** Foster teamwork among healthcare professionals.
4. **Evidence-based practice:** Use the best available research evidence to guide care.
5. **Transparency:** Share data, results, and lessons learned.

Quality Improvement Process:



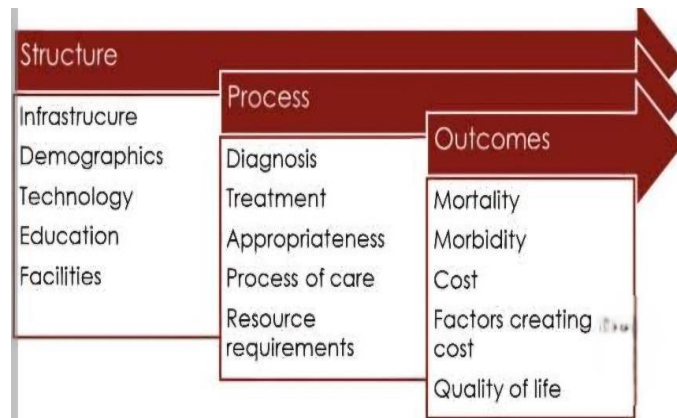
Role of Data Collection and Analysis in Quality Improvement

1. **Identify areas for improvement:** Analyze data to pinpoint gaps in care.
2. **Set goals and objectives:** Use data to establish realistic targets.
3. **Monitor progress:** Track data to assess effectiveness of interventions.
4. **Evaluate outcomes:** Analyze data to determine impact on patient care.
5. **Inform decision-making:** Use data to guide policy and practice changes.

Quality Improvement Models

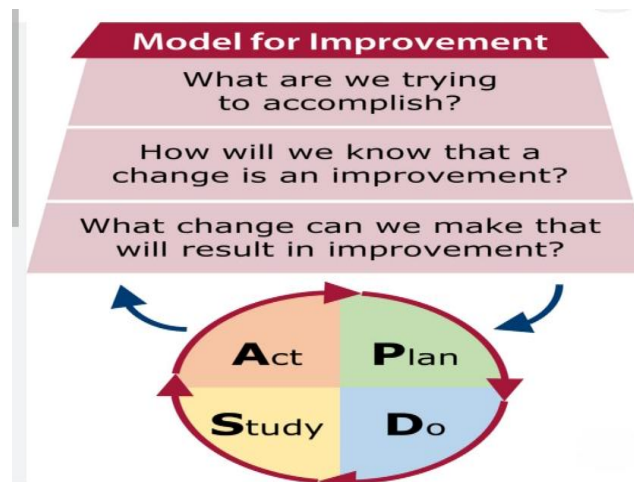
1. Donabedian model:

Focus on structure,
process, and
outcomes



AI-Enhanced Donabedian's
...Framework: A 21st-Century

IHI Model for Improvement: Use PDSA cycle to improve. A structured approach to testing changes.



Model for Improvement

3. Lean healthcare

model: Apply lean principles to healthcare. Lean principles: Eliminate waste, optimize workflows.



Start with a quality improvement (QI) model. Quality improvement models are frameworks that provide systematic, structured approaches to system or process improvement. While working with any QI model, the key is to carefully choose strategies that have the best chance to improve how your team interacts with patients.



4 Six Sigma model

Lean healthcare model: Apply lean principles to healthcare. Lean model was adapted to healthcare to improve efficiency, reduce waste, and enhance patient care. ---Core Idea of Lean:” **Do more with less**” — meaning deliver higher quality care using fewer resources, less time, and minimal waste, all while maximizing value for the patient.

Key Principles of Lean in Healthcare:

- 1 **Value:** Define what is truly valuable from the patient's perspective (e.g., timely care, safety, comfort).
2. **Value Stream:** Map out every step in a process (like patient admission or lab testing) to identify value-added and non-value-added activities.
3. **Flow:** Ensure that care processes flow smoothly without delays or bottlenecks.

4. **Pull:** Provide services only when needed, rather than overproducing or duplicating efforts.
5. **Perfection:** Continuously seek to improve and eliminate waste.



Types of Waste in Lean Healthcare (called "Muda"):

1. **Defects:** Medical errors, misdiagnoses.
2. **Overproduction:** Ordering unnecessary tests.
3. **Waiting:** Patients waiting for appointments or test results.
4. **Non-utilized talent:** Underusing staff skills.
5. **Transportation:** Moving patients unnecessarily.
6. **Inventory:** Overstocking medications or supplies.
7. **Motion:** Staff walking long distances to get tools.
8. **Extra processing:** Repeating documentation.

4.Six Segma Model for quality improvement



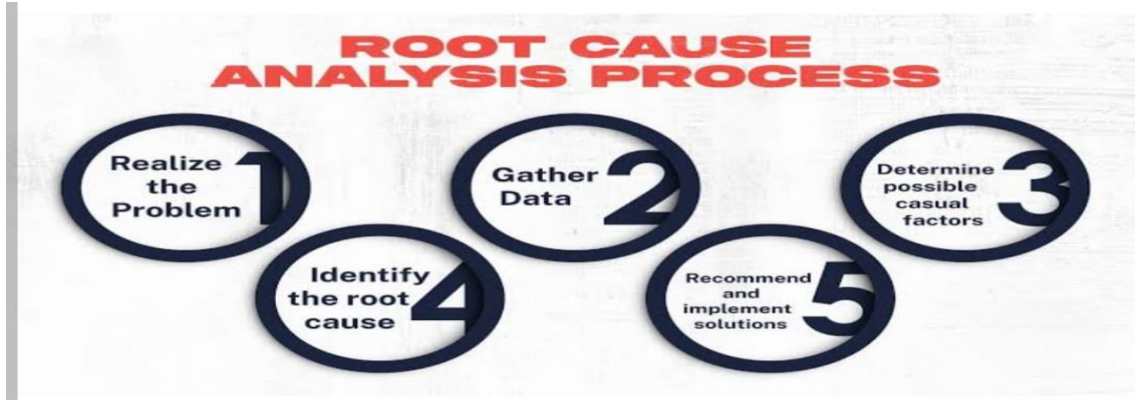
Quality Improvement Tools:

Quality Improvement Tools: is the process that helps better understanding, analyzing and communicate, quality improvement efforts.

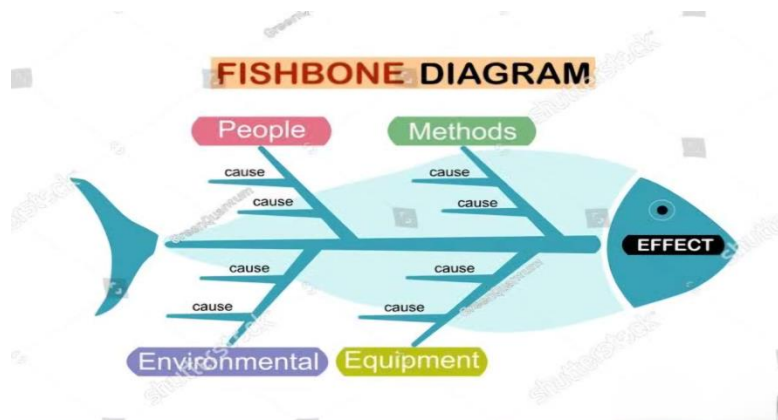
- **Root cause analysis (RCA).** (Fishbone diagram, 5 Whys Technique, Cause and Effect Diagram).
- **Failure mode and effects analysis (FMEA).**
- **SWOT Analysis.**
- **Flowcharts.**
- **Control charts.**
- **Pareto chart.**
- **Run charts.**

- **Decision Tree Analysis.**

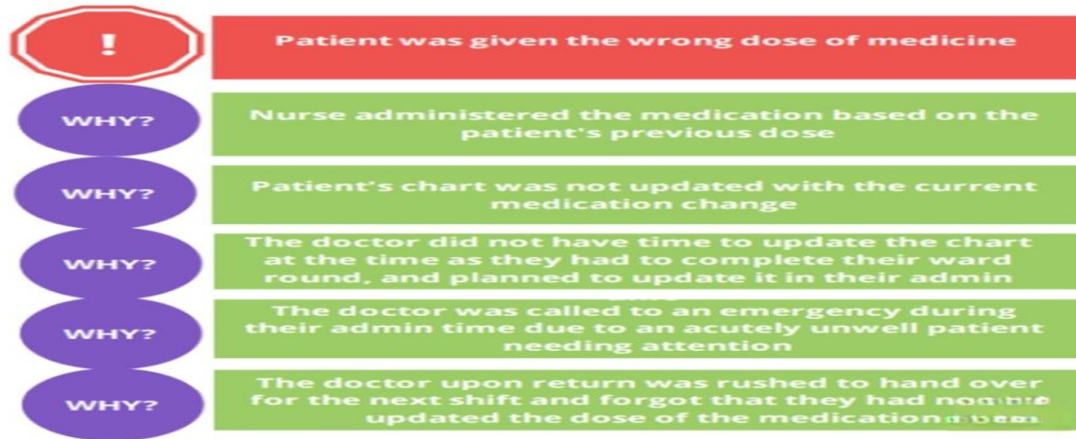
1. Root cause analysis (RCA): Identify underlying causes of problems.



A. Fishbone diagram: Visualize potential causes of problems. also calle **Ishikawa** Diagram one of (RCA) tools Visualize potential causes of problems.



B. 5 Whys Technique : depends on Asking Why 5 times to Identify the root cause of the problem.



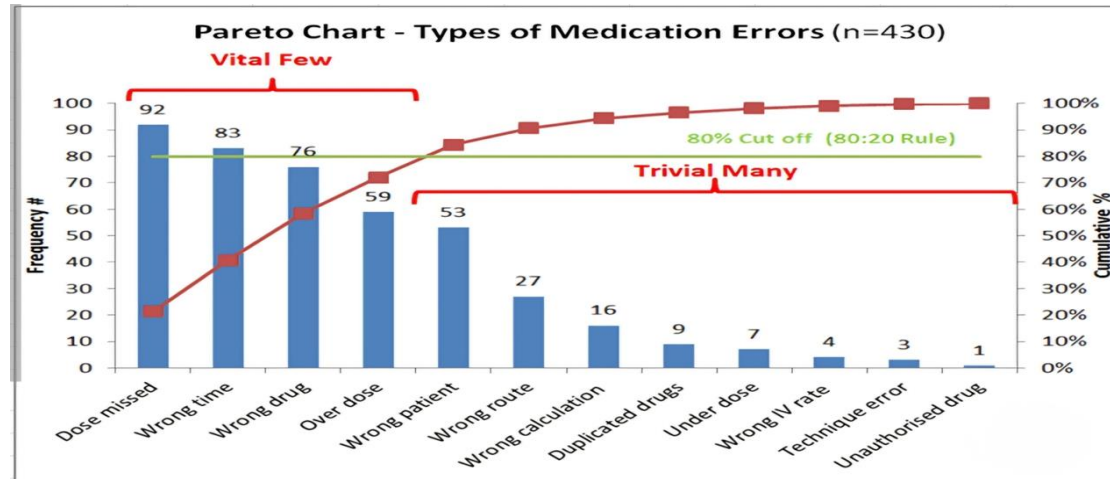
2. Failure mode and effects analysis (FMEA): Identify potential failures and their impact.



3. SWOT analysis: Identify strengths, weaknesses, opportunities, and threats.



4. Pareto chart: graph that helps identify the most significant factors contributing to a problem. It is based on the Pareto Principle (80/20 rule) — meaning that 80% of problems often come from 20% of causes.



5. Flowcharts: Map care processes to identify inefficiencies.
6. Control charts: Monitor data to detect trends and anomalies.
7. Run charts: Visualize data to track progress over time.
8. Shewhart chart.

Challenges and Opportunities for quality improvement:

1. Data quality and availability: Ensure accurate, reliable data.
2. Stakeholder engagement: Involve patients, families, healthcare professionals.
3. Sustainability: Ensure long-term viability of quality improvement initiatives.
4. Spread and scale: Disseminate successful initiatives across organization.



lecture questions:

Define the following:

- Quality Assurance.
- Quality Improvement
- Quality Control.
- Lean model.
- **List :**
- Tools of quality Improvement.
- Models of quality Management.

Source of waste according to lean model

Nursing Informatics for Quality Improvement:

Definition of Health Data: “Health data refers to data related to a person’s health; their health conditions, information relating to maternity and children, causes of death, and quality of life. Health data includes, for example: patient health records, studies about the health of groups of people, data from blood or tissue samples.

Health informatics: is a multidisciplinary field that includes computer science, information science, and healthcare. It focuses on the use of information technology to improve the quality of healthcare.

Nursing informatics:

Nursing informatics (Information Technology (IT) in Nursing) refers to the refers to the application of computer systems and software to improve patient care, enhance effectiveness, sufficiency, and make informed decisions in the field of nursing.

It involves the use of technology to manage and process health information, communicate with patients and healthcare providers, and support clinical decision-making

Advantages of Nursing Informatics:

There are many advantages for nursing informatics including the following:

- Support for their mission to deliver high quality, evidence-based care.
- Privacy, confidentiality, and security of health information.
- Promote improvement in key relationships with physicians, peers, patients, families, and interdisciplinary team members.
- Decreased time spent on documentation.



- Systematically collect data that can be transformed into information for decision making.
- Save Time and effort spent in Paper storage printouts.

Disadvantages of Nursing Informatics: -

1. Computer malfunction.
2. Impersonal effect: by making information available to anyone.
3. Concern for privacy: information can conceivably be accessed or destroyed by unauthorized people.
4. Dissemination of inaccurate information.
5. Limitation of standard vocabulary: Standardized and limited vocabulary or phrase structure in computer formats can be a restriction on accuracy or completeness of charting.
6. It will be many years until health care organization can move to paperless chart. Save time for patient care.
7. Cost: cost of the hardware and software, training, supervisory involvement during the transitional period of change, with accompanying work pace and procedure disruption.

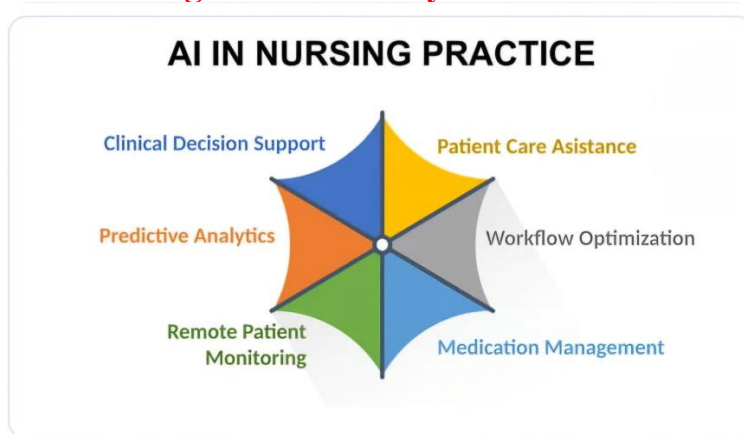
Role of informatics nurse: -

1. Improve nursing practice and the quality of patient care.
2. Facilitation of data collection for research.
3. Support decision based on assessments to drive a new plane of care.
4. Support for cost saving and productivity goals.
5. Creating new tools for patient care from new technologies
6. Integrating systems
7. Evaluating the effects of nursing systems.

Barriers to nursing informatics:

- Limited financial resources, as no budget is enough to buy computers and other machines.
- Resistance to change as many years' nurses use paper in documentation very difficult for them when they are old to use electronic devices
- Needs training and education to the nurses to know how to use new system or devices.
- Fear of computerization, most nurses are afraid to make mistakes while using advanced machines or computers.
- Time management, nurses need more time to accommodate new devices, and they will face an issue regarding time management.

Types of Nursing information system:



4. Electronic Medical Record (EMR):

An electronic version of traditional record used by health care providers. It includes information about a patient's health history, such as diagnoses, medicines, tests, allergies, immunizations, and treatment plans.

5. Electronic Health Records (EHRs):

- ▶ Digital versions of paper charts
- ▶ Store patient medical history, medications, allergies, immunization records, and more.

Which can

- ▶ It can be accessed from several different locations simultaneously, as well as by different levels of providers.
- ▶ Decision support tools as well as alerts and reminders notify the clinician of possible concerns or omissions. An example of this is the documentation of the patient allergies in the computer system. Health care providers would be alerted to any discrepancies in patient medication orders.
- ▶ Facilitate data collection for research: electronically stored client records provide quick access to clinical data for many clients.

6. Clinical Decision Support Systems (CDSS):

- ▶ Computer programs that analyze patient data and provide evidence-based recommendations.
- ▶ Help nurses make informed decisions about diagnosis and treatment plans.
- ▶ Reduce medical errors and improve patient outcomes.



7. Telehealth and Telemedicine:

- ▶ Use technology to deliver healthcare services remotely
- ▶ Enable virtual consultations, remote monitoring, and telemedicine appointments
- ▶ Expand access to care, especially for rural and underserved populations

8. **Coaching:** through using AI in training and educating nursing staff.

9. Wearable Technology:

- ▶ Devices that track various health metrics, such as heart rate, blood pressure, and activity levels
- ▶ Provide real-time data to healthcare providers and patients
- ▶ Promote preventive care and self-management .



10. Big Data Analytics:

- ▶ Analyze large datasets to identify patterns and trends Predict potential health risks and optimize resource allocation.
- ▶ Improve population health management.

1- Robots in health care system (AMR, Modular Robots and Surgical-Assistance Robots)

Robots in the medical field are playing an important role. Intel offers a diverse portfolio of technology for the development of medical robots, including surgical-assistance, modular, and autonomous mobile robots, which streamline supply delivery and disinfection,

transforming how surgeries are performed and enabling providers to focus on engaging with and caring for patients.

Types of Robotics in Healthcare

Using **robotics** in the medical field enables a high level of patient care, efficient processes in clinical settings, and a safe environment for patients and healthcare workers.

A. Autonomous mobile robots (AMRs).

High-Quality Patient Care .

1. Medical robots support minimally invasive procedures, customized and frequent monitoring for patients with chronic diseases.
2. In addition, robots alleviate nursing & caregivers workloads.

Streamlined Clinical Workflows

3. Simplify routine tasks, reduce the physical demands on human workers, and ensure more consistent processes.
4. These robots can address staffing shortages.
5. help make sure supplies, equipment, and medication are in stock where they are needed.
6. Cleaning and disinfection AMRs enable hospital rooms to be sanitized and ready for incoming patients quickly, allowing workers to focus on patient-centric, value-driven work.



Safe Work Environment

1. To help keep healthcare workers safe, AMRs are used to transport supplies and linens in hospitals where pathogen exposure is a risk.
2. Cleaning and disinfection robots limit pathogen exposure while helping reduce hospital acquired infections (HAIs)—and hundreds of healthcare facilities are already using them.
3. Social robots, a type of AMR, also help with heavy lifting, such as moving beds or patients, which reduces physical strain on healthcare workers.

B. Surgical-Assistance Robots

As motion control technologies have advanced, surgical-assistance robots have become more precise.

1. These robots help surgeons achieve new levels of speed and accuracy while performing complex operations with AI- and computer vision–capable technologies.
2. Some surgical robots may even be able to complete tasks autonomously, allowing surgeons to oversee procedures from a console.
3. Simulation platforms use AI and virtual reality to provide surgical robotics training. Within the virtual environment, surgeons can practice procedures and hone skills using robotics controls.

C. Modular Robots

Modular robots enhance other systems and can be configured to perform multiple functions.

1. In healthcare, these include therapeutic exoskeleton robots and prosthetic robotic arms and legs.

2. Therapeutic robots can help with rehabilitation after strokes, paralysis, or traumatic brain injuries or with impairments caused by multiple sclerosis.
3. A wheelchair-mounted robotic arm currently being developed by Intel aims to assist patients with spinal injuries in performing daily tasks.



Applying Autonomous Mobile Robots to Real-World. Some robots can assist professionals before a patient is checked in. For example, one autonomous robot in Mexico, developed by start-up Roomie, is being used to help medical staff with high-risk COVID-19 patients. Named Roomie Bot, it triages patients by taking their temperature, blood oxygen level, and medical history when they arrive at the hospital. Roomie Bot uses Intel-based technology.

The Future of Robotics and Healthcare

Health robotics will continue to evolve alongside advancements in machine learning, data analytics, computer vision, and other technologies. Robots of all types will continue to evolve to complete tasks autonomously, efficiently, and accurately.



lecture questions:

Define the following:

- Nursing informatics.
- Electronic Health Records.
- Electronic Medical Record (EMR)
- **List:**
- Types of Nursing information system.
- Advantages of Electronic Health Records: □
- Barriers to nursing informatics.
- Benefits of Robotics in Healthcare
- Role of informatics nurse.

References

1. Institute for Healthcare Improvement (IHI). (n.d.). Quality Improvement. Retrieved from <https://www.ihl.org/topics/quality-improvement/Pages/default.aspx>

2. Agency for Healthcare Research and Quality (AHRQ). (n.d.). Quality Improvement. Retrieved from <https://www.ahrq.gov/topics/quality-improvement/index.html>

3. Joint Commission. (n.d.). Quality Improvement. Retrieved from <https://www.jointcommission.org/standards-information/quality-improvement/>

▪ Patient Safety

1. World Health Organization (WHO). (2019). Patient Safety. Retrieved from <https://www.who.int/patientsafety/en/>



2. Agency for Healthcare Research and Quality (AHRQ).

(n.d.). Patient Safety. Retrieved from <https://www.ahrq.gov/topics/patient-safety/index.html>

3. The Joint Commission. (n.d.). Patient Safety. Retrieved from

<<https://www.jointcommission.org/standards-information/patient-safety/>'

- Risk Management

1. American Society for Healthcare Risk Management (ASHRM). (n.d.).

Risk Management. Retrieved from <https://www.ashrm.org/resources/risk-management/>

2. Agency for Healthcare Research and Quality (AHRQ). (n.d.). Risk

Management. Retrieved from <https://www.ahrq.gov/topics/risk-management/index.html>

3. The Joint Commission. (n.d.). Risk Management. Retrieved from

<<https://www.jointcommission.org/standards-information/risk-management/>'

- Leadership and Management

1. American College of Healthcare Executives (ACHE). (n.d.). Leadership

Development. Retrieved from <https://www.ache.org/leadership-development/>

2. American Organization for Nursing Leadership (AONL). (n.d.).

Leadership Development. Retrieved from <https://www.aonl.org/leadership-development>

3. Institute for Healthcare Improvement (IHI). (n.d.). Leadership. Retrieved from <https://www.ihi.org/topics/leadership/Pages/default.aspx>

- Communication and Teamwork



1. American Nurses Association (ANA). (n.d.). Communication and Teamwork. Retrieved from <https://www.nursingworld.org/practice-policy/health-policy/communication-and-teamwork/>
 2. Institute for Healthcare Improvement (IHI). (n.d.). Communication and Teamwork. Retrieved from <https://www.ihl.org/topics/communication-and-teamwork/Pages/default.aspx>
 3. Agency for Healthcare Research and Quality (AHRQ). (n.d.). Communication and Teamwork. Retrieved from <https://www.ahrq.gov/topics/communication-and-teamwork/index.html>
- Evidence-Based Practice
 1. Agency for Healthcare Research and Quality (AHRQ). (n.d.). Evidence-Based Practice. Retrieved from <https://www.ahrq.gov/topics/evidence-based-practice/index.html>
 2. Institute of Medicine (IOM). (2001). Crossing the Quality Chasm: A New Health System for the 21st Century.
 3. World Health Organization (WHO). (2019). Evidence-Based Practice. Retrieved from <https://www.who.int/evidence-based-practice/en/>